

Current Situation: Dropout Crisis in India

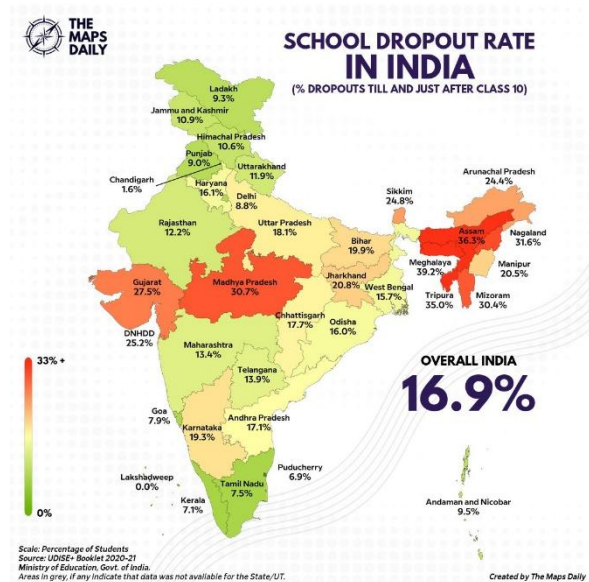
India's school system supports nearly 25 crore students, but the scale of dropout remains a persistent national crisis. Despite big strides, over 5.4 million students drop out every year, with secondary and higher-secondary stages hit the hardest.

- In Rajasthan, nearly 9 lakh (900,000) children left school in 2023-24 - translating to more than 2,400 students per day. [source](#)
- Elementary-level dropouts (2022–2024): A report analyzing UDISE+ data found that 899,240 children in Rajasthan dropped out at the elementary level (Grades 1–8) between the 2022–2023 and 2023–2024 academic years.
- Girls in Rajasthan face unique barriers: the dropout rate for girls can reach over 20% at age 16 and exceeds the national average at every stage. [source](#)
- Major causes: early marriage, family finances, too few mentors, migration, and lack of role models, especially for rural and SC/ST communities.
- Student dropouts remain a critical challenge in India's higher education. According to the **All India Survey on Higher Education (AISHE 2021–22, Ministry of Education)**, dropout rates in undergraduate programs range between **20–25%**, with even higher rates in rural and low-income groups.
The primary factors identified are:
 - Poor attendance and academic performance.
 - Financial hardships and unpaid fees.
 - Lack of timely interventions or counseling.
 - Limited parent awareness and engagement.
- Currently, institutes maintain **attendance, exam marks, and fee records in separate spreadsheets** or ERP silos. This fragmentation prevents early detection of students struggling on

multiple fronts simultaneously. Most interventions happen *after* term-end results, when disengaged students are already lost.

Why It Matters

- Each dropout has a ripple effect, affecting family livelihoods, future workforce strength, and even perpetuating cycles of early marriage and poverty.
- Studies show that for every year of secondary education missed, lifetime income potential can drop by ₹60,000-₹1,20,000 per person.



Existing Gaps: Where Traditional Approaches Fall Short

- Schools and government programs focus heavily on enrolment but lag when it comes to monitoring and acting on at-risk students.
- Critical risk factors (attendance dips, sudden marks drop, prolonged fee dues) are often missed due to fragmented data and manual processes.
- Mentors often lack actionable, evidence-based insight, especially in rural and resource-limited schools.

Our Solution: MentorAid – AI-Driven Dropout Prevention and Student Success Platform

MentorAid is a holistic, AI-powered solution designed to predict, prevent, and reduce student dropout at scale—especially in challenging environments like Rajasthan.

1. Integrated Data Intelligence :

MentorAid seamlessly combines student data from attendance, marks, fee payments, and institutional ERP/spreadsheet systems. Multi-source upload support means all relevant risk factors are captured—no more manual compilation or missing students.

2. AI-Powered Early Warning :

Our platform uses advanced machine learning models to analyze patterns, identify at-risk students early, and provide a transparent “risk score” with clear root causes. Unlike black-box systems, MentorAid offers explainable AI insights, so mentors and administrators see not just the warning, but “why” a student is vulnerable.

3. Personalized, Actionable Interventions :

MentorAid does not stop at prediction. It auto-generates tailored intervention plans, including:

- Draft emails and communication templates for parents and guardians.
- Syllabus Simplifier for converting complex topics into student-friendly lessons.
- Curated resources, remedial content, and recommendations specific to each student.
- Progress goals and tracking to ensure interventions are working and adjust as needed.

4. Role-Based Transparent Dashboards

A dynamic dashboard for every stakeholder:

- Mentors: Full risk and progress history, multi-file uploads, interventions, and AI-assist features.
- Administrators: Macro dashboards showing institute-, class-, or district-wide trends and bottlenecks for resource targeting.
- Students and Parents: Personalized progress, support tips, gamified engagement, and actionable feedback loops.

5. Inclusive, Scalable, and Accessible

MentorAid is designed for the field:

- Works online and offline, critical in remote or low-infrastructure schools.
- Delivers support and communication in multiple local languages.
- Built for fast, low-overhead deployment in both government and private institutions.

6. Future-Ready Features

The roadmap includes:

- Student sentiment and emotional distress detection via advanced NLP.
- Real-time financial counseling for at-risk families (with links to government schemes).
- Mentor gamification to boost proactive engagement.
- Deeper multi-language and accessibility support.
- Macro View – National Dashboard , the Institutions can compare with Regional and state level Performance.
- Offline first design as in many areas with internet issue, we use PWA (Progressive web app), which allows dashboard work offline and Sync when connected.

Our Impact and Projected Results

- If implemented across high-risk Rajasthan blocks:
 - Projected 30–40% reduction in secondary dropout within two academic years, especially for girls and SC/ST students.
 - Administrative time spent on dropout tracking cut by up to 70%.
 - Mentor engagement and intervention rates expected to double.
- Long-term: Socio-economic uplift - reduced early marriage, higher graduation rates, better financial resilience for families.
- Data-Driven Decision Making : Converts fragmented spreadsheets into actionable dashboards, empowering mentors, institutions, and policymakers with real-time insights.

- **Inclusive Student Support** : Hybrid parent communication (email, SMS, IVR) + multi-language support ensures accessibility across literacy levels and rural regions.
- **Institutional & Policy Alignment** : Provides institute-level efficiency and government-ready macro dashboards aligned with **NEP 2020** and **Digital India Vision 2030**.
- **Scalable & Sustainable Innovation** : Built on open-source, modular architecture → cost-effective, scalable across colleges, districts, and states without heavy infrastructure needs.

Future Roadmap

- **Emotion/Sentiment AI** to flag students with high stress or emotional risk, for full well-being support.
- **Financial Counseling Integration** to guide at-risk families to schemes and scholarships before financial crisis causes dropout.
- **Global Language Support** to expand MentorAid across geographies and underserved linguistic groups.
- **Gamified Mentor & Student Engagement** - Rewards, leaderboards, and recognition badges for mentors and students, Encourages consistent mentorship and motivates students with achievable milestones.
- **Predictive Analytics 2.0** - Advanced ML models (Deep Learning, Sequential Pattern Recognition), Behavioral pattern mining and dropout prediction timing optimization.

Conclusion

MentorAid is not just a tech solution—it's a social catalyst, built to address real pain points in Indian education. By focusing on Rajasthan as a “living lab” of the problem, and designing with scalability in mind, MentorAid is ready for real-world, measurable impact.

Additional Informative images :

